Dvara E-Registry: KhetScore & Doordrishti

Alternative Farm Underwriting for Smallholder Farmers Without Land Deeds

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1. The Problem

~85% Excluded

- **Exclusion:** Around 85% of farmers in India work small plots without formal land titles or credit files.
- **Collateral:** Banks demand land ownership papers, cutting off roughly 100 million farming households from fair loans.
- **Debt Trap:** Families rely on local moneylenders who charge heavy interest (around 40% to 50%).

2. The Product & How It Works

Crop Scoring

- **Concept:** Scores farmers on actual crop productivity instead of paper land ownership.
- **Satellite Data:** Uses public satellite feeds to track crop greenness, soil moisture, and seasonal patterns over time.
- **Simple Score:** Turns historical farm health into a straightforward credit score for partner banks.
- **Mobile App:** Field staff walk farm borders with a basic mobile tool to verify field locations on the ground.

3. The Investment Ask

Expansion

$2.0M
Funding Ask

Expanding the rural field agent team and growing into 5 more farming states.

- **Field:** Equip around 500 local agents with basic mobile kits to map fields.
- **Offline:** Build offline storage so the mobile tool works in remote dead zones.
- **Scale:** Connect about 250,000 small farmers to fair bank loans.

4. UN Sustainable Development Goals

UN Goals

- **SDG 1 (No Poverty):** Opens access to fair bank credit for families without land deeds.
- **SDG 8 (Decent Work & Growth):** Helps regional banks lend safely across rural communities.
- **SDG 9 (Innovation & Infrastructure):** Uses digital crop tracking to lower borrowing barriers for small growers.

Sources: ITU AI for Good (aiforgood.itu.int) | UN Sustainable Development Goals (sdgs.un.org) | Dvara E-Registry (dvaraeregistry.com)

Real-World Evidence & Performance Baseline

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Track Record Across Company Operations, Partner Banks, and Independent Field Studies

1. Company Track Record

Company Data

~50k+
Farmers Reached
Across 10 agricultural states

- **Loans:** Unlocked around $6M in fair bank credit for smallholder families.
- **Speed:** Cut loan approval times from about a month down to 2 to 3 days.
- **Mapping:** Georeferenced over 100,000 acres of small farm plots.

2. Partner Results

Lenders & FPOs

~2%
Portfolio Default Rate
Across partner bank portfolios

- **Repayment:** Partner banks report default rates staying around 2% or less.
- **Insurance:** Sends automatic crop insurance payouts when satellites confirm heavy flood damage.
- **Local Groups:** Works with farmer groups to help members buy seeds in bulk.

3. Independent Field Study

Field Trial

~40%
Credit Uptake
Higher loan adoption

~25%
Empowerment
Female autonomy gain

- **Trial:** Field evaluation run across rural farming villages in Odisha, India.
- **Adoption:** Documented roughly 40% higher formal loan uptake among small tenant farmers.
- **Earnings:** Verified steady gains in seasonal farm profits.
- **Decision Making:** Found about 25% higher female input on family farm spending choices.

CS 5083 Graduate Benchmark:
3x Lower Defaults
Compared to Normal Farm Bank Loans

- **National Baseline (RBI):** Central bank reports show standard agricultural loan default rates in India hover around 9% to 10%.
- **KhetScore Results:** Portfolios underwritten through satellite crop scoring maintain defaults near 2%, showing that tracking crop health is roughly 3x to 4x safer than traditional paper deed requirements.

Sources: IFPRI & 3ie Evaluation Study (ifpri.org) | Reserve Bank of India Baseline (rbi.org.in) | ITU AI for Good (aiforgood.itu.int)